Does the Channel Matter?

Separating Attention from Belief Correction in Parental Information Interventions

Jordi Torres Vallverdú

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Information Problems in Education

Students and parents make decisions under imperfect information

- Students hold biased beliefs about returns to education and own ability
  - Major choice and labor market outcomes [Arcidiacono et al., Wiswall and Zafar, 2015]
  - Persistence and dropout decisions [Stinebrickner and Stinebrickner, 2012, Arcidiacono et al., 2016]
  - College application and matching [Larroucau and Rios, 2026]
- **Parents are also misinformed** about their children's performance
  - Mean belief–performance gap > 1 SD in some contexts [Dizon-Ross, 2019]
  - Biases concentrated among low-SES households
- Parental investments respond to information $\Rightarrow$ misperceptions distort human capital accumulation

Information Treatments to Parents

Providing information to parents improves outcomes

- Dizon-Ross [2019]: Correcting beliefs about child ability in Malawi
  - Parents reallocate investments toward lower-performing children
  - Beliefs are systematically upward-biased
- Bergman [2021]: SMS-based grade notifications in the US
  - Parents receive same info via SMS vs. standard report cards
  - Result: 0.19 SD GPA increase at very low cost
  - $\sim$59% from belief revision, $\sim$41% from reduced monitoring costs
- **The puzzle:** the *channel* of delivery matters, not just content

Why Does the Channel Matter? Possible Mechanisms

Standard economics: only content should matter $\Rightarrow$ **three mechanisms**

1. **Biased beliefs & asymmetric updating**
   - Parents hold upward-biased priors; update asymmetrically
   - More information $\Rightarrow$ belief correction (upward)
2. **Limited attention & salience** $\leftarrow$ *main focus*
   - Under cognitive load, "child performance" drops out of active decision model [Gabaix, 2014]
   - SMS creates high salience contrast vs. report card among paperwork [Bordalo et al., 2012]
3. **Capacity to act & budget constraints**
   - Parents intend to engage but can't due to budget constraints.

Problem: existing designs bundle content and channel $\Rightarrow$ cannot separate mechanisms

Research Question

Core question: Holding timing fixed, does the *channel* of parental information delivery affect investment—and through which mechanism?

Bergman (2021): SMS-based grade info raises GPA by 0.19 SD. Is this because parents **learn something new**, or because they finally **notice** what they already could have known?

This design separates:

1. **Channel effect** (SMS vs. paper, content fixed) $\Rightarrow$ attention / salience
2. **Content effect** (detailed vs. standard, channel fixed) $\Rightarrow$ belief correction
3. **Interaction:** are channel and content complements or substitutes?

Key design choice: all arms receive information *per-term* $\Rightarrow$ timing and action set equalized

Model: Production Function & Belief Updating

Grade production: Student i in school e:

$$Y_{it}^{e} = Y(a_i,\, e_{it}^{p},\, e_t^{s}) + \varepsilon_{it}$$

- a_i: student ability $\quad e_{it}^{p}$: parental effort $\quad e_t^{s}$: school effort

Belief updating: Parents hold prior $\hat{a}_{it}^{\text{pre}}$, receive signal σ_{it}:

$$\hat{a}_{it}^{\text{post}} = \hat{a}_{it}^{\text{pre}} + \underbrace{A(\kappa_i)}_{\text{attention}} \cdot \underbrace{\lambda(\sigma_{it}, \hat{a}_{it}^{\text{pre}})}_{\text{salience}} \cdot (\sigma_{it} - \hat{a}_{it}^{\text{pre}})$$

- $A(\kappa_i) \in [0, 1]$: attention weight [Gabaix, 2014]
  - High cognitive load $\kappa_i \Rightarrow$ low attention; SMS lowers κ_i
- $\lambda(\cdot) \in [0, 1]$: salience weight [Bordalo et al., 2012]
  - Signals contrasting with priors get more weight

Model: Effort Choice

Parental effort: Given updated beliefs, parent maximizes:

$$\max_{e_{it}^p \geq 0} \ \ln Y(a_i,\, e_{it}^p,\, e_t^s)\Big|_{\hat{a}_i^{\text{post}}} - C^p(e_{it}^p,\, z_i)$$

- $C^p(e_{it}^p, z_i)$: cost of effort, heterogeneous in parent type z_i (SES, time)
- Allows credit constraints to enter through cost function

Key distinction:

- $A(\kappa_i)$: determines whether parent *considers* the information at all
- $\lambda(\cdot)$: determines how much weight the signal gets *conditional on attention*

$\Rightarrow$ Both needed: a parent can attend to a signal yet underweight it (or vice versa)

The RCT: Factorial Design

Setting: Primary/secondary schools with digital gradebooks but low baseline parent–school communication. Multiple terms per year (e.g. 3 trimesters). Randomization stratified by school.

2×2**:** all arms deliver info at **same frequency** (biweekly, mid-term):

	Standard content	**Detailed content**
Paper (mailed home)	C	T2
SMS / WhatsApp	T1	T3

- **T4**: Same as T3, weekly frequency $\Rightarrow$ sustained attention vs. one-shot

Panel structure: observe grades and effort across terms $\Rightarrow$ trace *dynamics* of belief updating and effort response within the model

Identification Strategy

2×2 **factorial** with equalized timing enables clean decomposition:

Estimating equation (panel across terms t):

$$Y_{it} = \alpha + \beta_1 \cdot \mathbf{1}[\text{SMS}]_i + \beta_2 \cdot \mathbf{1}[\text{Detailed}]_i + \beta_3 \cdot (\text{SMS} \times \text{Detailed})_i + \delta_t + X_i'\gamma + \epsilon_{it}$$

- β_1: **channel / attention effect** A — content and timing held fixed
- β_2: **content / belief-correction effect** through $\lambda \cdot (\sigma - \hat{a})$
- β_3: **complementarity** between attention and beliefs
- δ_t: term fixed effects — capture common shocks across periods

Dynamics: with multiple terms, can test whether effects are *immediate* (attention) or *cumulative* (belief correction builds over repeated signals)

Additional: elicit baseline beliefs $\hat{a}_i^{\text{pre}}$ to separate biased priors from distorted updating

Power Calculations

Benchmark: Bergman [2021] joint effect = 0.19 SD. Individual mechanisms smaller ⇒ **MDE = 0.10 SD**.

Assumptions: $\alpha = 0.05$, power $= 0.80$, ICC $\rho = 0.05$, cluster size $m = 20$, $\sigma = 1$.

Required sample per arm (clustered):

$$n_{\text{per arm}} = \underbrace{\frac{(z_{\alpha/2} + z_\beta)^2 \cdot 2\sigma^2}{\text{MDE}^2}}_{\approx 1{,}568} \times \underbrace{[1 + (m-1)\rho]}_{\text{DEFF} = 1.95} \approx 3{,}058$$

Factorial efficiency: each main effect pools two cells (e.g. T1+T3 vs. C+T2):

Total sample	$\approx 6{,}000$	students
Per cell	$\approx 1{,}500$	students
Clusters	≈ 300	schools/classes? (75/cell)

Conclusion & Policy Implications

Summary:

- Existing evidence shows information treatments work, but bundles channel and content
- 2×2 factorial with equalized timing cleanly separates attention from belief correction
- Panel structure across terms traces the dynamics of parental response

The mechanism determines the policy:

$\beta_1 \gg \beta_2$	Attention drives effect $\Rightarrow$ invest in **delivery design** (SMS, nudges)
$\beta_2 \gg \beta_1$	Beliefs drive effect $\Rightarrow$ **richer content** suffices, channel secondary
$\beta_3 > 0$	Complements $\Rightarrow$ **both channel and content** needed together

Contribution:

- First design to separate channel from content in parental information interventions
- Low-cost, scalable — directly informs how schools communicate with parents
- Structural model to extend results beyond local effects.

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